# Subject Description Form

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| **Subject Code** | ABCT2101 |
| **Subject Title** | Biochemistry |
| **Credit Value** | 3 |
| **Level** | 2 |
| **Pre-requisite** | General Chemistry I, General Biology |
| **Co-requisite** | Nil |
| **Exclusion** | Introductory Cell Biology and Biochemistry<br>(The subject “Biochemistry” and “Introductory Cell Biology and Biochemistry” are mutually exclusive of the other) |
| **Objectives** | The aims of this subject are for students to acquire a basic understanding of common biomolecules such as carbohydrates, lipids, amino acids and enzymes; and to appreciate the importance of their unique structure and biochemical reactions involved. |
| **Intended Learning Outcomes** | Upon completion of the subject, students will be able to:<br>a. recognize the structure and properties of simple carbohydrates, lipids, and amino acids; and relate the structure and function of various important biomolecules.<br>b. understand the basic principles of carbohydrate and lipid metabolism to appreciate how energy is being preserved, and extracted and utilized in biological systems.<br>c. explain the essential principles of enzymology and solve problems in enzyme kinetics and mechanisms.<br>d. apply the basic biochemical techniques on enzyme characterization and metabolite assays and interpret and analyze biochemical data.<br>e. develop analytical, critical thinking, and written communication skills. |
| **Subject Synopsis/ Indicative Syllabus** | Carbohydrate & lipids: structure, properties and functions (6 hours)<br>Amino acids, peptides and proteins (6 hours)<br>Case studies of proteins: structure-function relationship (6 hours)<br>Basic principles of enzymology: enzyme kinetics and mechanism (6 hours)<br>Bioenergetics: glycolysis, TCA cycle, beta-oxidation and OXPHOS (15 hours) |
| **Teaching/Learning Methodology** | Lectures are designed to provide students with the basic concepts of structure-function relationship of common biomolecules and the principles of metabolism and enzymology. To enhance their learning and knowledge, problem-based |

learning approach will be adopted. Students will be given assignments in some topics for further exploration in depth to gain thorough understanding. Tutorial classes and Blackboard platform will be used to gauge their learning and performance. A variety of assessment tools will be used, including quizzes, assignments, and the final exam to develop students' analytical skills, critical thinking and communication skills.

**Assessment Methods in Alignment with Intended Learning Outcomes**

| Specific assessment methods/tasks | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) | | | | |
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| | | a | b | c | d | e |
| 1. Attendance | 10 | √ | √ | √ | | |
| 2. Assignments | 10 | √ | √ | √ | √ | √ |
| 3. Quizzes | 30 | √ | √ | √ | √ | √ |
| 4. Examination | 50 | √ | √ | √ | √ | √ |
| Total | 100 % | | | | | |

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Assignments, quizzes and examination are used to gauge how much students have learned in the structure-function relationship of different common biomolecules, basic metabolism and enzymology. Writing skills will be assessed in all the assessment tasks and methods. The laboratories and laboratory reports in particular demand students to demonstrate their competence in executing biochemical assays and in the interpretation and analysis of experimental data.

Students are required to attend at least 75% of scheduled sessions for the subject. Students fail to fulfill the attendance requirement will lose the 10% attendance score and not be eligible to register ABCT3108.

**Student Study Effort Expected**

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| Class contact: | |
| ▪ Lecture | 26 Hrs. |
| ▪ Tutorial | 13 Hrs. |
| Other student study effort: | |
| ▪ Assignment | 10 Hrs. |

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| | ▪ Self study | 61 Hrs. |
| | Total student study effort | 110 Hrs |
| **Reading List and References** | **<u>Essential</u>**<br>Nelson, D. L. and Cox, M.M.<br>Lehninger Principles of Biochemistry, 2021 | |